Homework 3:

Calling Mutations with Samtools/Bcftools

Since I have already aligned the reads in Homework 2, the next step is to call mutations (variants) using Samtools and Bcftools. Below is a high-level overview of the steps involved.

Step 1: Download the Reference Genome (hg19)

To perform mutation calling, you need to download the hg19 reference genome. Run the following command in your terminal:

wget http://hgdownload.cse.ucsc.edu/goldenpath/hg19/bigZips/hg19.fa

Step 2: Generating Variant Calls in VCF Format Using Bcftools

Once the `hg19.fa` reference genome and the aligned, sorted BAM file (`ENCFF002DKA_DKE_Aligned_Sorted.bam`) are available, you can call variants using Bcftools.

1. Generate Raw SNP Calls

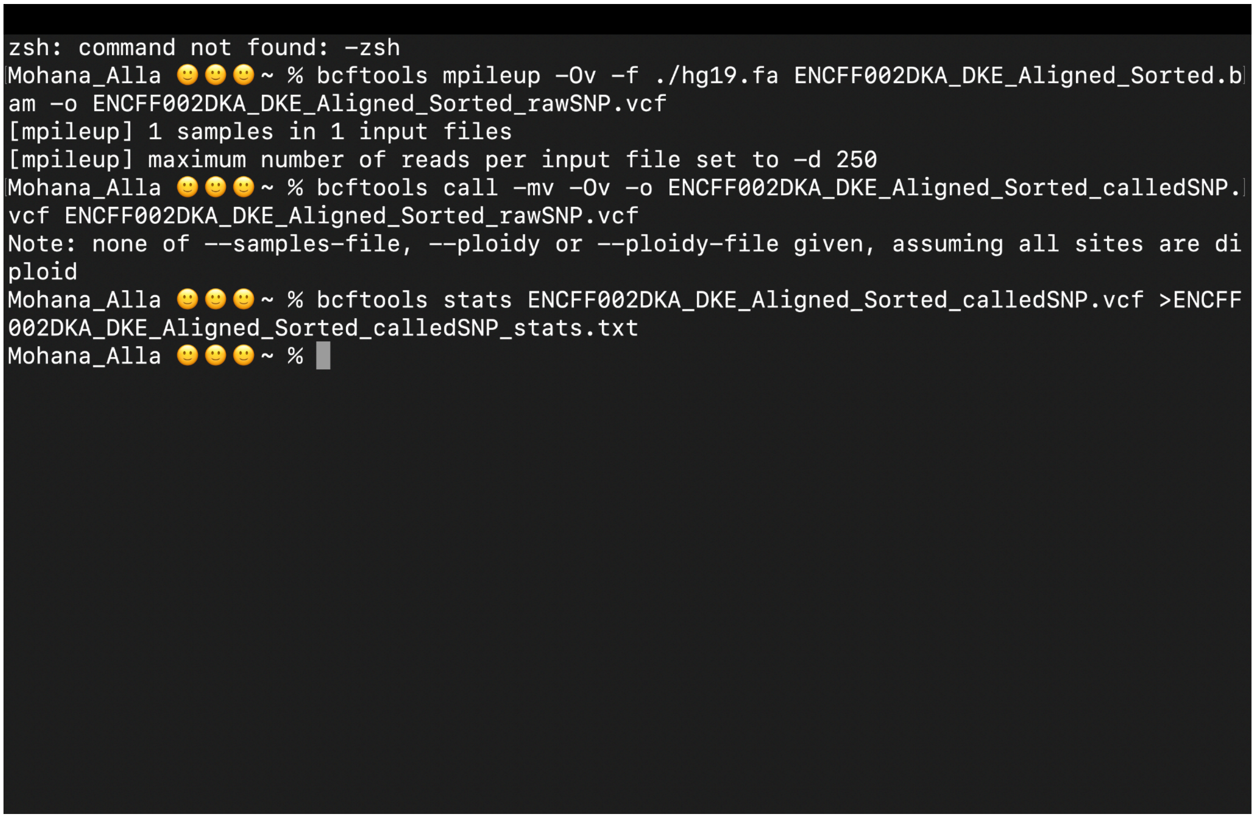
Run the following command to generate raw SNP calls:

bcftools mpileup -Ov -f ./hg19.fa ENCFF002DKA_DKE_Aligned_Sorted.bam -o ENCFF002DKA_DKE_Aligned_Sorted_rawSNP.vcf

Explanation of the command:

- `bcftools mpileup`: Generates a pileup of reads at each position.
- `-O v`: Outputs the file in VCF format (instead of BCF).
- `-f ./hg19.fa`: Specifies the reference genome.
- `ENCFF002DKA_DKE_Aligned_Sorted.bam`: The aligned and sorted BAM file.

- `-o ENCFF002DKA_DKE_Aligned_Sorted_rawSNP.vcf`: Outputs the raw SNP calls to this file.



2. Call Variants Using Bcftools

After generating the raw pileup, call actual mutations using:

bcftools call -mv -Ov -o ENCFF002DKA_DKE_Aligned_Sorted_variants.vcf ENCFF002DKA_DKE_Aligned_Sorted_rawSNP.vcf

Explanation:

- `bcftools call`: Calls SNPs and Indels.
- `-m`: Uses the multiallelic caller.
- `-v`: Outputs variants only (excludes sites with no variation).
- `-O v`: Outputs the file in VCF format.
- `-o ENCFF002DKA_DKE_Aligned_Sorted_variants.vcf`: The final variant file.

Step 3: Analyzing Variant Statistics Using Bcftools

Once the VCF file (`ENCFF002DKA_DKE_Aligned_Sorted_calledSNP.vcf`) is generated, analyze the basic statistics of the variants.

1. Running Bcftools Stats

Run the following command to generate variant statistics:

bcftools stats ENCFF002DKA_DKE_Aligned_Sorted_calledSNP.vcf > ENCFF002DKA_DKE_Aligned_Sorted_calledSNP_stats.txt

Explanation:

- `bcftools stats`: Generates summary statistics of the VCF file.
- `ENCFF002DKA_DKE_Aligned_Sorted_calledSNP.vcf`: Input VCF file containing called variants.
- `> ENCFF002DKA_DKE_Aligned_Sorted_calledSNP_stats.txt`: Redirects output to a text file.

Step 4: Visualizing the HBD Gene Region in IGV

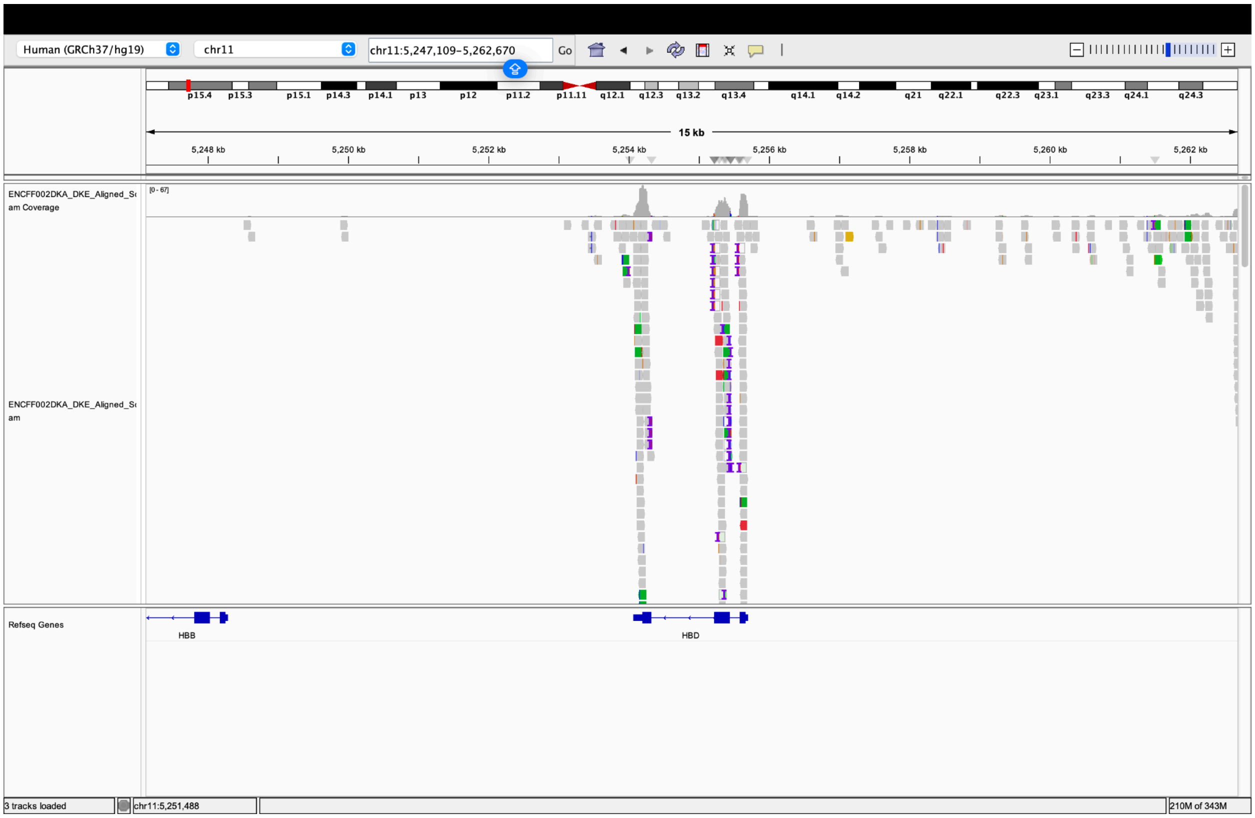
To visualize the HBD gene region in your aligned data, follow these steps:

1. Load the BAM File into IGV

- Open IGV and click **"File"** → **"Load from File"**.
- Select the BAM file: `ENCFF002DKA_DKE_Aligned_Sorted.bam`.

2. Navigate to the HBD Gene Region

- In the IGV search bar, enter:
  HBD
- Press **Enter** to visualize the region.



Detailed Summary of Mutation Results

1. Variant Counts and Distribution

- **Total Variants Identified**: 1,112,892
- **SNPs (Single Nucleotide Polymorphisms)**: 944,664 (~85% of total variants)
- **Indels (Insertions/Deletions)**: 168,228 (~15% of total variants)
- **Multiallelic Sites**: 5,466
- **Multiallelic SNP Sites**: 118

Interpretation:

The majority of detected variants are SNPs, which is expected in genomic mutation analysis. The presence of multiallelic sites suggests some regions contain multiple alternate alleles.

2. Transition/Transversion Ratio (Ts/Tv)

- **Transitions (Ts)**: 626,807

- **Transversions (Tv)**: 317,975
- **Ts/Tv Ratio**: 1.97

Interpretation:

The Ts/Tv ratio is close to 2, which is normal for human genomes. Transitions (A↔G, C↔T) occur more frequently than transversions (A↔C, A↔T, C↔G, G↔T) due to mutation bias and selective pressure.

3. Allele Frequency (AF) Distribution

- **Singleton Mutations (AF = 0.000)**: 308,439 SNPs
- **High-Frequency Variants (AF = 0.99)**: 636,343 SNPs

Interpretation:

A large number of singleton variants suggests many rare mutations in the dataset. The presence of high-frequency variants (AF = 0.99) indicates that some SNPs are common polymorphisms found in the population.

4. Variant Quality Distribution

- **Quality scores range**: 3 to >90
- **Majority of variants**: Quality scores between 3 and 40

Interpretation:

Many variants have low-quality scores, indicating the need for filtering (`QUAL ≥ 20`) to remove low-confidence mutations. A higher quality threshold ensures that only reliable variants are retained for downstream analysis.

Recommendations and Next Steps

1. Filtering Variants

- Remove low-quality variants using a threshold of `QUAL ≥ 20` and `DP ≥ 10` for reliable calls.

2. Variant Annotation

- Use tools such as **ANNOVAR, SnpEff, or VEP** to predict the functional impact of variants (e.g., missense/nonsense mutations).

3. Visualization in IGV

- Load aligned BAM files into IGV and inspect key gene regions (e.g., **HBD gene**) to validate mutations.

4. Further Statistical Analysis

- Compare identified SNPs and Indels with known disease-associated mutations and population databases.

Final Conclusion

This mutation analysis reveals a high number of SNPs, a Ts/Tv ratio consistent with human genomes, and a mix of rare and common variants. To improve dataset reliability, filtering, annotation, and IGV visualization are recommended to identify biologically significant mutations.